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MATEMATISKA INSTITUTIONEN, STOCKHOLMS UNIVERSITET

Gröbner Bases and Elimination in Macaulay 2

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Abstract

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1 Introduction

Hello world
[DAC10]
[LM21]

2 The Algebreic Abstraction of Geometry

Algebra is the mathematical expression of the human need for structure and abstraction. Through algebra, humans try to capture more of the mathematical, and natural, domain under their pen. The endless hunger for knowledge can express itself in various way. When using algebra as a tool, creating abstract models that capture generic structure is favored. However, these abstract models can be difficult to understand. The ability to navigate between levels of abstraction is a vital tool for any mathematician interested in applying algebreic methods.

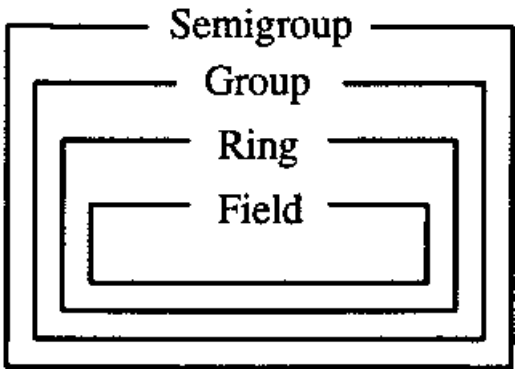
This paper assumes a degree of familiarity with algebreic structures, but the uninitiated peruser is kindly directed towards the easy-going introduction of "Introducetion to Modern Algebra" [Wei70] with bite size exercises, or alternativly, the more rigorous "Abstract Algebra" [DF03] for a deeper understanding.

As the section header suggests, this section will be primarily focused with defining the tools necessary to build an analytic bridge between geometry and algebra so that we can analyze geometric objects with algebra. A function that maps geometric objects to algebreic must be established. This function uses Affine Varietes and Ideals, along with the dictionary established in the Hilbert Strong Nullstellensatz, to accomplish that goal.

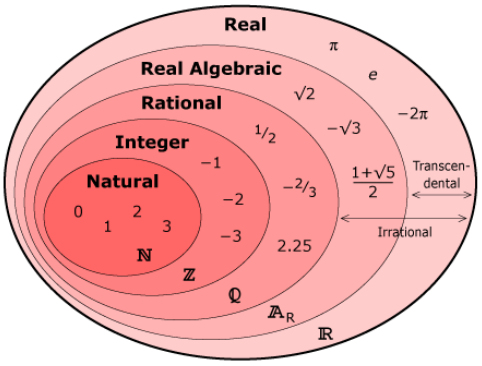
2.1 Levels of Abstraction

Abstract algebra traverses through layers of abstraction as an exercise. As an example, if we are interested in how groups act when supplemented with commutivity (Abelian groups), it would be useful to start at groups and traverse the layers of figure 1a downwards. Note that climbing upwards may not be as fruitful for this particular case.

Choosing an Abelian group that stays beteween groups and rings is $(\mathbb{R}, +)$. Here the set of real numbers is only considered with the binary operator $+$, and commutivity follows. Analyzing this group and all the properties that are associated with it is of interest. However, much use can be extracted from climbing further down the abstraction ladder and analyzing rings a. Keeping it concrete with numbers, one can consider the abelian ring $\mathbb{Z}$. Time can be spent on this layer of abstraction and knowledge can be obtained through leveraging $\mathbb{Z}$'s special properties. Going even further down the ladder, to a field, one could analyze $\mathbb{Q}$. At each layer, new intuition can be gained. Note also, that while traversing downwards in the group abstraction,



(a) Abstractions within Groups [Lat]



(b) Generalization of Numbers [Diand]

Figure 1: Traversing the ladder of Abstraction

we were climbing up and down in the generalization of numbers. Starting at the top with $\mathbb{R}$, we jumped all the way down to $\mathbb{Z}$, and then up again to $\mathbb{Q}$. It is important to consider this, as different layers of abstraction do not neccessarily have to correspond with another. Caution is required when dealing with this iterative abstract process.

2.2 Polynomials

In order to adress the problem at hand fully, we need to generalize the idea of a polynomial. This is done by compressing more and more inforamation into a single mathematical expression. A process of abstraction is necessary. We start with the monomial, continue to a polynomial, and finally consider polynomial rings. All work done is done on these polynomials rings, and this process of abstraction enables us to say something about all monomials simultaneously.

Definition 2.1. A **monomial** in $x_1, \ldots, x_n$ is a product of the form

$$x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n} \tag{1}$$

where all $\alpha_1, ..., \alpha_n$ are nonnegative integers.

As the root word "mono" suggests, there is only one of these terms x^α.However, as we are creating the tools to generalize and deal with more abstract concepts, we also want to include the ability to capture any linear combination of these monomials.

Definition 2.2. A **polynomial** f in $x_1, \ldots, x_n$ with coefficients, $a_1, \ldots, a_n$, in a

field $\mathbb{K}$ is a finite linear combination of monomials.

$$f = \sum_{\alpha} a_\alpha x^\alpha, \quad a_\alpha \in \mathbb{K} \tag{2}$$

With the polynomial definitions at hand, we can construct the final abstraction for our mathematics involving polynomials, $\mathbb{K}[x] = \mathbb{K}[x_1, \ldots, x_n]$. An important note is that $\mathbb{K}[x]$ satisfies the axioms for rings and is therefore refered to as the *polynomial ring* When leveraging this powerful abstraction, we capture n dimensions of variables on any field $\mathbb{K}$. This enables more compact analysis.

2.3 Affine Spaces and Varieties

To start the bridge between the world of Algebra as we know it and geometry, there must be an established way to discuss geometry algebreically. This is where the concepts of affine spaces comes in.

Also talk about what geometry actually is in this context (points and lines and functions and so on)

This concept allows the analysor to view the geometry algebreically. However, this is not the endgoal, but an analytic step between geometry and the more powerful algebreic tools that we want to apply to geometry.

Relate also how vector spaces and affine spaces use maps.

" The geometric properties of a vector space are invariant under the group of bijective linear maps, whereas the geometric properties of an affine space are invariant under the group of bijective affine maps, and these two groups are not isomorphic." [Gal25]

Affine maps, crudely speaking, are generalizations of linear maps [Gal25]. This generalization can be viewed in different ways, but a few important examples are listed.

1. Affine spaces allow for a dynamic origin. Vector spaces do not.

2. There are more affine maps than there are linear maps.

3. Affine spaces study points in a more general matter, independently of the coordinate system chosen. Frame invariance.

2.3.1 Affine Spaces

In order to understand affine varieties, affine spaces must be understood. This coordinate-agnostic space is of interest since more abstract enteties can be modeled upon it. Let us first consider the definition:

Definition 2.3. An *affine space* is either the empty set, or a triple $< E, \vec{E}, + >$. E is a nonempty set of points while $\vec{E}$ is a vector space (of translations). The action $+$ is an operation on both E and $\vec{E}$ defined as follows: $+ : E \times \vec{E} \to E$. The following axioms are assumed for affine space:

A1 $a + 0 = a$ for every $a \in E$.

A2 $(a + u) + v = a + (u + v)$ for every $a \in E$ and every $u, v \in \vec{E}$.

A3 For any two points $a, b \in E$, there is a unique $u \in \vec{E}$ such that $a + u = b$

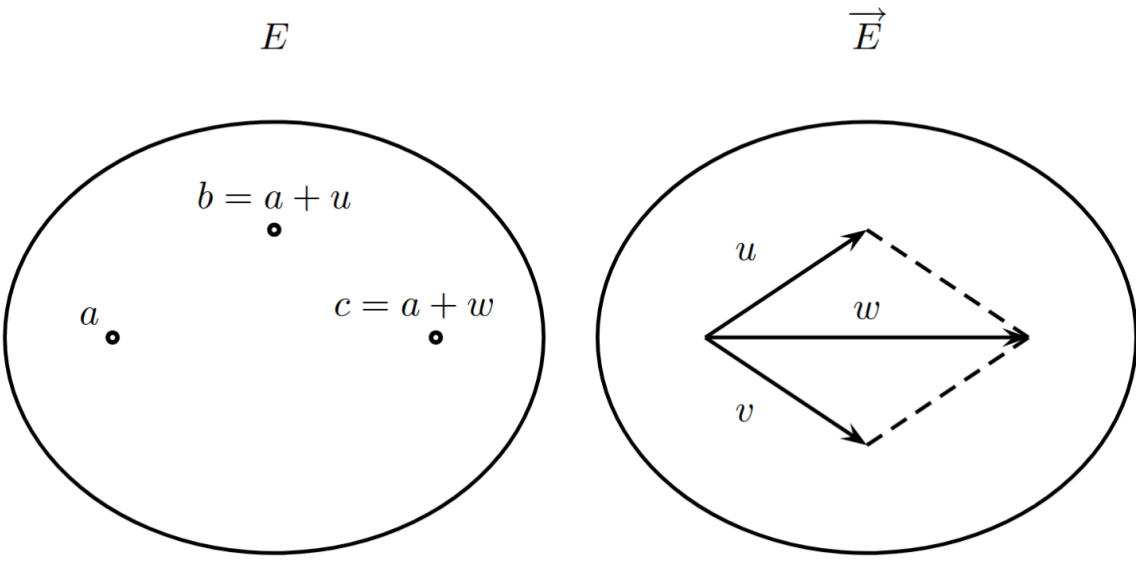


Figure 2: Intuiative understanding of affine space [Gal25]

The definition is a bit stuffy, but the intuition should be clear when considering figure 2. Here a seperation between the vectors and the points *to which the vector is applied to* should be noted. These two can be defined independently. This seperation is exactly what enabled coordinate-agnostic property that is of interest.

A supplemental definition will be provided for the affine to clear up any confusion.

Definition 2.4. Given a field $\mathbb{K}$ and a positive integer n, define the $n-$*dimensional affine space* over $\mathbb{K}$ as:

$$\mathbb{K}^n := \left\{[\hat{x}_1, \ldots, \hat{x}_n]^T : \hat{x}_i \in \mathbb{K}, i = 1, \ldots, n\right\} \tag{3}$$

This supplement elucidates the uncertainty behind how this space is defined as all other spaces. It's vector space has translations and they are restricted to the field upon which they operate.

2.3.2 Polynomials in Affine Spaces

It is time to relate the previously discussed polynomial to affine spaces. Leveraging definition 2.4, we can relate the two. [LM21] refers to this as a "key idea" and shows that the polynomial $p \in \mathbb{K}[x]$ yields a polynomial *function* as follows:

$$p : \mathbb{K}^n \to \mathbb{K} \tag{4}$$

The function maps the multidimensional field onto the monodimension. This p takes any $\hat{x} \in \mathbb{K}$ and associates a unique $p(\hat{x})$ to it. This is done easily by substituting $x = \hat{x}$ in the polynomial p. One important note is relevant to the zeros of p when taking $\mathbb{K}$ as a infinite field (e.g. $\mathbb{Z}, \mathbb{R}$). Then p is the zero polynomial if and only if $p(\hat{x}) = 0$ for all $\hat{x} \in \mathbb{K}$.

2.3.3 Affine Varieties

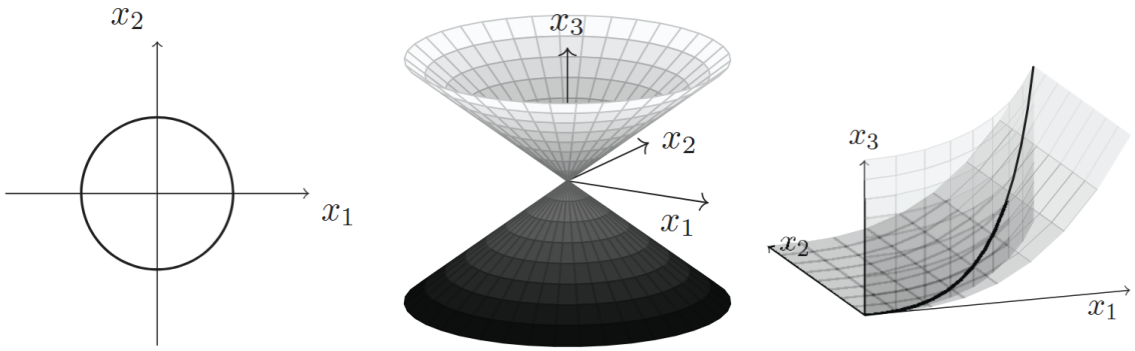
With an understanding of how the space we will operate upon will work, the first algebreic structure of geometry will be discussed: affine varieties.

Definition 2.5. Let $\mathbb{K}$ be a field, and let $f_1, \ldots, f_s$ be polynomials in $\mathbb{K}[x_1, \ldots, x_n]$. Then, set

$$\boldsymbol{V}(f_1, \ldots, f_s) = \{(a_1, \ldots, a_n) \in \mathbb{K}^n | \ f_i(a_1, \ldots, a_n) = 0 \ \forall 1 \leq i \leq s\} \tag{5}$$

We call $\boldsymbol{V}(f_1, \ldots, f_s)$ the **affine variety** defined by $f_1, \ldots, f_s$.

In other words, the affine varity is the set of $(a_1, \ldots, a_n)$ that solve all the equations in the variety simulteniously. Even though $\boldsymbol{V}$ might simply look like a graphed equation, figure 3



(a) $\mathbf{V}(x_1^2 + x_2^2 - 1)$. (b) $\mathbf{V}(x_1^2 + x_2^2 - x_3^2)$. (c) $\mathbf{V}(x_1^2 - x_2, x_1^3 - x_3)$.

Figure 3: Exmaples of affine spaces [LM21]

2.4 Rings and Ideals

2.5 Ordering Polynomials

2.6 An Analytic Bridge

2.7 Hilbert Strong Nullstellensatz

3 Gröbner Bases

3.1 Hilbert Bases

3.2 Gröbner Bases

3.3 Properties of Gröbner Bases

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